📄 **Project Documentation**

Retail Customer Segmentation using RFM Analysis (Python)

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1. Introduction

In today's competitive retail environment, understanding customer behavior is essential for improving customer retention and increasing revenue. Businesses collect large amounts of transactional data but often lack structured analysis to extract meaningful insights.

This project focuses on **Retail Sales Data Analysis** using **RFM (Recency, Frequency, Monetary) Analysis** to segment customers based on purchasing behavior and provide actionable business insights.

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2. Objective of the Project

The main objectives of this project are:

- To analyze customer purchasing behavior using retail transaction data
- To calculate Recency, Frequency, and Monetary values for each customer
- To segment customers into meaningful groups
- To support business decision-making through data visualization

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3. Problem Statement

Retail companies face challenges such as:

- Identifying high-value customers
- Designing targeted marketing strategies
- Increasing repeat purchases

This project addresses these challenges by applying RFM analysis to customer transaction data.

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4. Dataset Description

Dataset Characteristics

- Number of rows: 1000
- Number of columns: 9
- Data type: Retail transaction data

Key Columns Used

Column Name	Description
Customer ID	Unique customer identifier
Invoice No	Transaction ID
Invoice Date	Date of purchase
Quantity	Number of items purchased
Unit Price	Price per item
Total Amount	Quantity × Unit Price

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5. Tools and Technologies Used

- **Programming Language:** Python
- **Libraries:**
    - Pandas – data manipulation
    - NumPy – numerical operations
    - Matplotlib – data visualization
    - Seaborn – statistical visualization
- **Environment: Google Collab**

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6. Methodology

6.1 Data Cleaning and Preprocessing

- Converted Invoice Date to datetime format
- Created a new column:
- Total Amount = Quantity × Unit Price

6.2 RFM Metric Calculation

A snapshot date was chosen one day after the last transaction date.

Recency

- Number of days since the customer's last purchase

Frequency

- Total number of transactions made by the customer

Monetary

- Total amount spent by the customer

Each metric was calculated per customer using group-by operations.

6.3 RFM Scoring

Customers were divided into quartiles (1–4):

- **Recency:** Lower recency → higher score
- **Frequency:** Higher frequency → higher score
- **Monetary:** Higher spending → higher score

Ranking was applied to Frequency to handle duplicate values.

6.4 Customer Segmentation

Customers were grouped into segments based on RFM scores:

- Champions - Purchased **very recently,** Purchase **frequently,** Spend **a lot of money**
- Loyal Customers - Purchase **regularly, spent moderately to high, Spend moderately to high**
- Potential Loyalists - Purchased recently, But not very frequent yet Moderate spending
- Lost Customers - Haven't purchased in long time, Low frequency, Low spending

This segmentation helps businesses target customers effectively.

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7. Data Visualization

The following visualizations were created:

- Recency, Frequency, Monetary Histograms
- RFM heatmap (Recency vs Frequency vs Monetary)
- Segment-wise revenue bar chart
- RFM score distribution chart
- Cumulative revenue

- Boxplots for segment comparison

These visuals support better understanding of customer behavior.

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8. Results and Insights

- Most customers made only one purchase, indicating low retention
- A small group of customers contributes significantly to total revenue
- Some customers show high recency values and are at risk of churn
- High-value customers can be targeted for loyalty programs

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9. Business Recommendations

- Introduce loyalty programs to encourage repeat purchases
- Provide personalized offers to high-value customers
- Re-engage inactive customers using targeted campaigns
- Focus marketing resources on profitable customer segments

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10. Conclusion

This project demonstrates how **RFM analysis** can be effectively used to segment customers and generate meaningful insights from retail data. The analysis supports data-driven marketing strategies and helps improve customer retention and revenue growth.

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11. Future Enhancements

- Apply machine learning clustering techniques
- Predict Customer Lifetime Value (CLV)
- Create an interactive dashboard (Power BI)
- Automate the RFM pipeline for real-time analysis

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12. Learning Outcomes

- Hands-on experience with real-world retail data
- Understanding of customer segmentation techniques

- Improved Python data analysis and visualization skills
- Ability to translate data insights into business actions